

Linguistic Distance

Thom Lazor

Mentor: Jan Šnajder

University of Zagreb – FER

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Introduction & Motivation

- Languages differ intuitively (e.g., English vs Croatian vs Ukrainian)
- Practical importance:
 - Language learning difficulty
 - Multilingual NLP performance
 - Policy in multilingual societies

Linguistic Subsystems

- **Phonetics:** speech sounds
- **Phonology:** sound systems and rules
- **Morphology:** word structure and inflection
- **Syntax:** sentence structure and order
- **Semantics:** meaning representation

Asymmetry in Distance

- True distances: symmetric, obey triangle inequality
- Linguistic distances often asymmetric:
 - Mutual intelligibility (e.g., Dutch $\rightarrow$ English $\neq$ English $\rightarrow$ Dutch)
 - Divergence-based metrics (e.g., KL Divergence)

Measuring Linguistic Distance

- **Lexical distance** (cognates, edit distance)
- **Phonemic distance** (phoneme inventories, phonemic edit distance)
- **Language acquisition** difficulty
- **Mutual intelligibility** (cloze tests, surveys)
- **Model-based divergence** (e.g., KL of language models)

Spectral Analysis

- Decomposes signals into frequencies
- Can reveal hidden patterns and structures
- Common in physics, engineering, music

Spectral Analysis of Speech

- Tools:
 - Short-Time Fourier Transform (STFT)
 - Wavelet Transform
 - Welch's Method for Power Spectral Density (PSD)
- Applications:
 - Language/dialect identification
 - Accent and prosody analysis
 - text to speech/speech to text

Analysis of Text

- Text as structured signals:
 - Character n-grams
 - POS tags, syntactic patterns
- Applications:
 - Style and authorship detection
 - Dialect identification
 - Structural analysis of genres

Key Datasets and Resources

- **WALS**: 2,662 languages, 192 features
- **ASJP**: 7,726 languages, normalized Levenshtein distance
- **XNLI**: 15-language inference corpus
- **Bible-Corpus**: Aligned texts in 100 languages
- **mBERT / XLM-R**: Multilingual transformer models

Conclusion

- Linguistic distance = multi-dimensional and messy
- Used in language education, NLP, typology, policy
- Spectral methods may offer objective, interpretable analysis